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A cumulative offer process for supply chain networks

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Abstract
We introduce a version of the Cumulative Offer Algorithm of Hatfield and Milgrom (Am Econ Rev 95(4):913–935, 2005) modified for the supply chain networks setting of Ostrovsky (Am Econ Rev 98(3):897–923, 2008). The algorithm provides an alternative proof for existence of a chain stable network. Moreover, we show that when choice functions satisfy same-side substitutes and cross-side complements, and the Law of Aggregate Demand and Supply, then contracts satisfy Irrelevance of Rejected Contracts condition, a condition implicitly assumed in Ostrovsky (2008).

Keywords Supply chain networks · Matching with contracts · Market design

JEL Classification C78 · D63 · D85 · L14

1 Introduction

The matching with contracts model of Hatfield and Milgrom (2005) is one of the most important contributions to the literature on two-sided matching. Hatfield and Milgrom (2005) provide a general and elegant framework that subsumes many of the previously separate specialized matching models—both with and without endogenously determined salaries, as well as more general relationships specified by a bilateral contract for each match. Moreover, it has led to various important applications in settings traditionally outside the scope of two-sided matching.

The supply chain networks model of Ostrovsky (2008) is a significant extension beyond the standard "two-sided" matching framework that considers matching on a network. Ostrovsky (2008) presents a novel approach to an industrial organization

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topic, using the tools of matching, and the result is a mathematically elegant and intuitive model of supply chains. He extends the two-sided matching notion of a pairwise stable allocation to that of a *chain stable network* for the supply chain setting.

Hatfield and Milgrom (2005) propose two methods to show the existence of a stable allocation in the two-sided, many-to-one matching with contracts setting: the generalized Gale-Shapely algorithm, which utilizes a fixed-point theorem, and the cumulative offer process, in which agents on one side propose and agents on the other side choose contracts from from a set of accumulated offers. The fixed-point approach in Hatfield and Milgrom (2005) generalizes Adachi (2000)'s use of Tarski's fixed-point theorem to prove existence of stable matchings in the setting of one-to-one matching. Echenique and Oviedo (2004, 2006) extend Adachi (2000)'s fixed-point approach to the settings of many-to-one and many-to-many matching, respectively.

Ostrovsky (2008)'s approach extends the fixed-point approach to the supply chain matching setting. Ostrovsky (2008) provides a natural generalization of substitutable preferences over contracts—same-side substitutability and cross-side complementarity—and shows that such preferences are sufficient for chain stable networks to exist. To prove existence, Ostrovsky (2008) introduces the T-algorithm, a fixed-point algorithm that finds a chain stable network.

In this paper, we present an extension of the cumulative offer process of Hatfield and Milgrom (2005), modified for the supply chain matching setting. Our Supply-Chain Cumulative Offer Algorithm (SCCOA) is an extension of the many-to-many cumulative offer process in Fung and Hsu (2014). We assume choice functions, rather than preferences, to be the primitives of the model. As Chambers and Yenmez (2017) argue, the concepts of stability and substitutability are more natural in a choice-theoretic setting. The SCCOA provides an alternative proof for existence of chain stable networks when choice functions satisfy same-side substitutability and cross-side complementarity. Moreover, the proof provides insight into the interactions that lead to chain stability amongst different tiers in a supply chain network.

Aygün and Sönmez (2013) and Aygün and Sönmez (2012b) show that Hatfield and Milgrom (2005) implicitly assume a condition called Irrelevance of Rejected Contracts (IRC) condition for the existence result.[1] Loosely speaking, the IRC condition means that rejecting a contract does not affect an agent's choice set. Following the example in Aygün and Sönmez (2012b), we show that the IRC condition is also implicitly assumed in Ostrovsky (2008). In our proof of existence, we indicate the place where the IRC condition is used. Finally, we show that when preferences satisfy Hatfield and Kominers (2012)'s Law of Aggregate Demand and Supply, in addition to same-side substitutability and cross-side complementarity, then choice functions satisfy the IRC condition.

Since Ostrovsky (2008)'s seminal contribution, matching in networks has been extended beyond supply chain matching. Hatfield and Kominers (2012) introduce bilateral contracts to generalize both the many-to-many and supply chain matching settings. Assuming preferences satisfy same-side substitutability and cross-side complementarity, which the authors term *full substitutability*, and acyclicity, which rules

[1] Aygün and Sönmez (2012a) further show the use of IRC in Hatfield and Kojima (2010), which shows existence of stable allocations under weakened substitutability condition in many-to-one matching with contracts.

out contract chains that form cycles, the existence of a stable network is established using a fixed-point approach. Note that Hatfield and Kominers (2012)'s definition of stability implies Ostrovsky (2008)'s chain stability, though they are equivalent under full substitutability and acyclicity. Thus, our SCCOA finds a stable network in the setting of matching with bilateral contracts under full substitutability and acyclicity.

Hatfield et al. (2013) generalize Hatfield and Kominers (2012) to a setting with continuous transfers and establish a connection between stable networks and competitive equilibria. Hatfield et al. (2018) show that in *general trading networks*, with full substitutes and Laws of Aggregate Demand and Supply (i.e., "mono-substitutable" choice functions), chain stability in this setting is equivalent to stability as defined in Hatfield et al. (2013) and Hatfield and Kominers (2012). This result subsumes the results in Hatfield and Kominers (2012) and Hatfield et al. (2013), implying that under mono-substitutability chain stability is equivalent to competitive equilibrium in general trading networks. Note that the definition of chain stability in Hatfield et al. (2018) is distinct from the definition in Ostrovsky (2008) as it is adapted to a more general setting.

Fleiner et al. (2018) introduce possibly cyclic trading networks, generalizing the trading networks of Hatfield et al. (2013, 2018) and Hatfield and Kominers (2012). Fleiner et al. (2018) introduce the *trail stability* solution concept and show that full substitutability and the IRC condition imply that trail stability is equivalent to chain stability. The proof relies on Tarski's fixed-point theorem. Adachi (2017) provides an alternative fixed-point algorithm to prove the existence of trail stable networks. The approach is a generalization of Ostrovsky (2008)'s T-algorithm.

Cumulative offer processes have received increased attention in non-standard matching applications. Hatfield and Kojima (2010) propose the *bilateral substitutes* (which is weaker than substitutes) and the *unilateral substitutes* (which is stronger than bilateral substitutes but weaker than substitutes) conditions, and use a cumulative offer processes to show the existence of a stable allocation in many-to-one matching with contracts when choice functions satisfy bilateral substitutes. Sönmez and Switzer (2013) use a cumulative offer algorithm in Army cadet-branch matching, in which substitutable preferences do not hold. Kominers and Sönmez (2016) use a cumulative offer mechanism in matching with "slot-specific priorities," which function as diversity constraints and impose a structure on preferences such that substitutability does not hold. Aygün and Turhan (2017) use a cumulative offer process in a school-choice setting with affirmative action constraints. In all of these settings, the absence of substitutability means fixed-point approaches do not work.

Studying the properties of cumulative offer processes has also attracted recent interest. Hirata and Kasuya (2014) show that the cumulative offer process is order independent for many-to-one matching with contracts, assuming choice functions satisfy bilateral substitutes and the IRC condition. Order-independence of the cumulative offer process in turn implies many agents making offers simultaneously is equivalent to each agent making offers sequentially. Afacan (2017) shows that for many-to-one matching with contracts, the cumulative offer process respects improvements under unilateral substitutes, IRC, and the Law of Aggregate Demand. Finally, Hatfield et al. (2017) provide necessary and sufficient conditions for a stable and strategy-proof

mechanism to exist in many-to-one matching with contracts and show that if a stable and strategy-proof mechanism exists, then it must be a cumulative offer mechanism.

In the next section, we present Ostrovsky (2008)'s model, as well as the IRC condition. Section 3 presents the cumulative offer algorithm and states the main results. In "Appendix A", we show that a stable allocation may not exist without the assumption of the IRC condition. In "Appendix B", we provide an example of the operation of the Cumulative Offer Algorithm. The proofs are in "Appendix C".

2 Model

There is a finite set of nodes A and a partial ordering $\succ$ on A such that, for $a, b \in A$:

1. $a \succ b$ means that b is a potential buyer for a and that a is a potential seller for b,
2. $a \not\succ b$ and $b \not\succ a$ means no relationship is possible for a and b.

The order $\succ$ determines the possible trading relationships for each pair of nodes, i.e., who is the upstream agent and who is the downstream agent, and so determines a supply chain structure on A. Note that transitivity rules out loops, e.g., a upstream for b and b upstream for c means a cannot be downstream for c.

There is a finite set of bilateral contracts $\mathcal{X} \equiv \{x = (s, b, l, p) : s \succ b, \text{ for } s, b \in A\}$, where $l \in \mathbf{N}$, $p \in \mathbf{R}$ are the terms of the contract between seller s and buyer b. Each contract specifies one unit l of a good between a buyer and seller at price p. Each buyer and seller may sign multiple contracts between themselves, and any number of contracts either as buyer or as seller. Signing two contracts that only differ in price is not feasible.

Given a contract $x \in \mathcal{X}$, let $A(x)$ denote the set of agents involved in x, let $B(x)$ denote the buyer involved in x, and $S(x)$ denote the seller involved in x.

For a set of contracts $X \subseteq \mathcal{X}$, let $A(X) \subseteq A$ denote the set of agents associated with contracts in X. Similarly, let $B(X)$ denote the set of buyers associated with X and let $S(X)$ denote the set of sellers associated with X. For each agent $a \in A$, let $X_a \subset X$ denote the set of contracts in X associated with agent a.

2.1 Choice functions

For each agent $a \in A$ and a set of contracts X, let $D_a(X) \equiv \{x \in X : S(x) = a\}$ be the set of contracts in X at which a is the seller. Similarly, let $U_a(X) \equiv \{x \in X : B(x) = a\}$ the set of contracts in X at which a is the buyer.

Given a set of contracts X, agent a's choice set $C_a(X) \subseteq X$ is a's chosen set of contracts from X. Thus, $D_a(C_a(X))$ and $U_a(C_a(X))$ are the downstream and upstream contracts chosen by a from X, respectively.[2]

An agent a's choice functions satisfy the following two properties:

Definition 1 (Ostrovsky 2008) Contracts satisfy **same-sided substitutability** if for two sets of contracts X and Y, the followings hold.

[2] Subscripts are omitted when it is clear the reference is to an agent $a \in A$.

1. $D(X) = D(Y)$ and $U(X) \subset U(Y) \Rightarrow U(X) \backslash U(C(X)) \subset U(Y) \backslash U(C(Y))$,
2. $U(X) = U(Y)$ and $D(X) \subset D(Y) \Rightarrow D(X) \backslash D(C(X)) \subset D(Y) \backslash D(C(Y))$.

The first part says that adding new contracts for a as a buyer (upstream) does not cause previously rejected upstream contracts to be chosen. The second part is analogous for adding new contracts for a as a seller. This is the standard notion of substitutability when there are only two sets of agents (i.e., two tiers), such that one set of agents are exclusively the sellers and the other set of agents are exclusively buyers.

Definition 2 (Ostrovsky 2008) Contracts satisfy **cross-sided complementarity** if for two sets of contracts X and Y, the followings hold.

1. $D(X) = D(Y)$ and $U(X) \subset U(Y) \Rightarrow D(C(X)) \subset D(C(Y))$,
2. $U(X) = U(Y)$ and $D(X) \subset D(Y) \Rightarrow U(C(X)) \subset U(C(Y))$.

The first part says that adding new contracts for a as a buyer does not cause previously chosen contracts for a as a seller to be rejected. The second part is analogous for adding new contracts for a as a seller. Thus, adding contracts on one side complements contracts on the other side. Hatfield and Kominers (2012) call preferences satisfying both conditions fully substitutable.

Finally, we assume the **Irrelevance of Rejected Contracts** (IRC) condition of Aygün and Sönmez (2013). Formally, contracts satisfy IRC for agent $a \in A$ if, for any set of contracts Y and a contract $z \in \mathcal{X} \backslash Y$,

$$z \notin C(Y \cup \{z\}) \Rightarrow C(Y) = C(Y \cup \{z\}).$$

That is, rejected contracts do not affect an agent's chosen set. Aygün and Sönmez (2012a, 2013) demonstrate that the IRC condition is important when choice functions, rather than preferences, are primitives of the model.

"Appendix A" presents an example in which chain stable allocations may not exist without the IRC condition. In the proof, we specify where the IRC condition is used.

2.2 Tiers

Node $a' \in A$ is an **ancestor** to node $a \in A$ if $a' \succ a$, that is, a' is upstream to a. Let $\overline{a} \equiv \{a' \in A : a' \succ a\}$ denote the set of ancestors to a. We partition A into K **tiers** $A = \cup_{i=1}^{K} A_i$, such that $A_i \cap A_j = \emptyset$, $i \neq j$, and two nodes $a, a' \in A$ are in the same tier A_i if

1. $a \not\succ a'$ and $a' \not\succ a$,
2. $|\overline{a}| = |\overline{a}'|$.

In other words, a, a' have no potential trading relationships, and have the same number of ancestors. Node a is in a **higher tier** than a' if $|\overline{a}| < |\overline{a}'|$. The highest tier A_1 is the set of nodes without ancestors, and the lowest tier A_K is the set with the most ancestors.

Given the partial ordering on A, tiers represent different stages in a supply chain. The highest tier consists of sellers with the most basic goods. Each tier below is

an intermediate step to the final good, which is sold to the lowest tier (e.g., end consumers). With this interpretation, it is natural to think of a process in which agents in the highest tier sign contracts with agents in the tier directly below (i.e., their potential customers), who evaluate such offers upstream based on potential trading relationships downstream, and so on.

2.3 Chain stability

A **network** μ is a set of contracts, with $\mu(a)$ denoting the set of contracts for a in network μ. Network μ is **individually rational** if $C_a(\mu(a)) = \mu(a), \forall a \in A$. For $n \geq 1$, a sequence of contracts $\mathbf{z} \equiv \{z_1, \ldots, z_n\}$ is called a **chain** if $b_i = s_{i+1}, \forall i < n$, where $b_i \equiv B(z_i)$ and $s_i \equiv S(z_i)$. A **chain block** for network μ is a chain $\mathbf{z}$ such that

1. $z_i \notin \mu, \forall z_i \in \mathbf{z}$,
2. $z_1 \in C_{s_1}(\mu(s_1) \cup z_1)$,
3. $z_n \in C_{b_n}(\mu(b_n) \cup z_n)$,
4. $\{z_i, z_{i+1}\} \subset C_{b_i}(\mu(b_i) \cup \{z_i, z_{i+1}\}), \forall i < n$.

A **chain stable network** is individually rational and does not have a chain block.

3 Main results

The following algorithm is a modification of the cumulative offer algorithm of Hatfield and Milgrom (2005), extended for many-to-many matching with contracts in Fung and Hsu (2014), and modified for the supply chain networks setting.

Supply Chain Cumulative Offer Algorithm (SCCOA)

Initialize $t := 1$ and define

- $O_a^U(1) = \emptyset$ for all $a \in A$.
- $O_a^D(1) = D_a(\mathcal{X})$ for all $a \in A$.

Step 1 An arbitrary agent $a' \in A$ offers

$$Y(t) = D_{a'}(C_{a'}(O_{a'}^U(t) \cup O_{a'}^D(t))).$$

Each agent $a \in A$ holds

$$H_a(t) = U_a(C_a(O_a^U(t)) \cup Y_a(t) \cup O_a^D(t)).$$

Step 2 If no proposals are made in Step 1, terminate the algorithm. Otherwise,

- Update $O_a^U(t+1) = O_a^U(t) \cup Y_a(t)$, for all $a \in A \backslash \{a'\}$.
- Update $O_a^D(t+1) = O_a^D(t) \backslash \{x \in O_a^D(t) : x \in O_{B(x)}^U(t+1) \backslash H_{B(x)}(t)\}$, for all $a \in A$.
- Set $t := t + 1$ and go back to Step 1.

Suppose the algorithm terminates at step T. The allocation produced by the algorithm is

$$\mu \equiv \bigcup_{a \in A} H_a(T) = \bigcup_{a \in A} U_a(C_a(O_a^U(T) \cup O_a^D(T))).$$

Example 1 in the Appendix illustrates the operation of the algorithm.

The algorithm belongs to the class of deferred acceptance algorithms, with agents making offers downstream (i.e., to buyers). We initialize the algorithm at $t = 1$ with a pair of opportunity sets for each agent $a \in A$. The upstream opportunity set, $O_a^U(t)$, represents all the available contracts at which a is the buyer. The downstream opportunity set, $O_a^D(t)$, represents all the available contracts at which a is the seller. Since offers are made downstream, upstream opportunity sets are initially empty.

The first step consists of a single agent making offers and agents responding to that agent's offer. An arbitrary agent $a' \in A$ makes an offer, $Y(t)$, which represents the downstream contracts from the set of contracts chosen by a' given his upstream and downstream opportunity sets. In response, every agent a evaluates $Y_a(t)$, the contracts in $Y(t)$ in which a is the buyer, together with their upstream and downstream opportunity sets, and holds $H_a(t)$, the set of upstream contracts chosen from this combination. Of course, if $Y_a(t) = \emptyset$ for an agent a, then a only evaluates contracts from his opportunity sets.

If no proposals are made, that is, there is no agent a' for which $Y(t)$ is non-empty, the algorithm is terminated. Otherwise, we update $t = t + 1$ and the opportunity sets and repeat until no more proposals are made. Upstream opportunity sets are updated by adding any contracts that were offered by a' at $t - 1$ to $O_a^U(t - 1)$. Downstream opportunity sets are updated by removing any contract in $O_a^D(t - 1)$ that was rejected in step $t - 1$. Thus, upstream opportunity sets are weakly larger and downstream opportunity sets are weakly smaller.

If the algorithm terminates at some step T, the final allocation is the set of all held contracts at step T. In other words, the final allocation is the set of all downstream offers that have not been rejected by buyers.

We are now ready to present the main results.

Theorem 1 *A chain stable network exists, when choice functions satisfy same-sided substitutes, cross-side complements, and the IRC condition.*

3.1 The Laws of Aggregate Demand and Supply

The Law of Aggregate of Demand is introduced in Hatfield and Milgrom (2005) to obtain the rural hospitals theorem and strategy-proofness in the many-to-one matching with contracts setting. The condition states that, when an agent is presented with more contracts, then he weakly chooses more. Aygün and Sönmez (2013) show that substitutes and the Law of Aggregate Demand imply IRC in the many-to-one matching with contracts setting.

Hatfield and Kominers (2012) propose the Laws of Aggregate Demand and Supply in the supply chain networks setting. Formally, let $X' \subseteq X'' \subseteq \mathcal{X}$. Choice functions

for node $a \in A$ satisfy the **Law of Aggregate Demand** if, for $D(X') = D(X'')$ and $U(X') \subseteq U(X'')$,

$$|U(C(X''))| - |U(C(X'))| \geq |D(C(X''))| - |D(C(X'))|.$$

Choice functions satisfy the **Law of Aggregate Supply** if, for $U(X') = U(X'')$ and $D(X') \subseteq D(X'')$,

$$|D(C(X''))| - |D(C(X'))| \geq |U(C(X''))| - |U(C(X'))|.$$

The Law of Aggregate Demand states that an agent facing more upstream contracts does not take on more downstream contracts than he does upstream contracts. Loosely speaking, agents commit to "buy more" by at least as much as they commit to "sell more." If an agent has no downstream contracts (for example, at the lowest tier), then we have Hatfield and Milgrom (2005)'s original Law of Aggregate Demand. The Law of Aggregate Supply, on the other hand, states that an agent facing more downstream contracts does not take on more upstream contracts than he does downstream contracts. That is, agents commit to "sell more" by at least as much as they commit to "buy more."

The following result recovers the IRC condition by assuming the Laws of Aggregate Demand and Supply, together with same-side substitutes and cross-side complements.[3]

Proposition 1 *If choice functions satisfy same-side substitutes and cross-side complements, and the Laws of Aggregate Demand and Supply, then contracts satisfy the IRC condition.*

4 Discussion

The supply chain matching framework is particularly useful for analyzing hierarchical organization structures, e.g., in creating a new division of a firm, and vertical production processes, e.g., wood in production of furniture. In a typical supply chain through which wood moves, the cumulative offer algorithm reflects how supply chain networks are formed since wood is key to production at every stage and securing downstream contracts may hinge on having upstream contracts.

The role of the IRC condition for existence has previously been noted by Blair (1988) and Echenique (2007). Aygün and Sönmez (2012a, 2013) illustrate it is implicitly assumed in two-sided matching with contracts. Our paper shows that the IRC condition is implicitly assumed in the more general supply chain networks setting.

[3] We thank an anonymous referee for shortening the proof.

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Compliance with ethical standards

Conflict of interest Both authors declare that they do not have any conflict of interest related to the research in this paper.

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Appendix A: The role of the IRC condition: an example

The following example is based on Example 1 in Aygün and Sönmez (2012b).

There are four nodes, $A = \{a_1, a_2, a_3, a_4\}$, with three tiers. Node a_1 has two contracts x, x'. Node a_2 has one contract y. Node a_4 has one contract z. Nodes a_1, a_2 and a_4 can only sign contracts with node a_3. Figure 1 illustrates all of the possible contracts that can be signed.

Consider the following choice functions:

$$C_{a_1}(\{x\}) = \{x\} \qquad C_{a_2}(\{y\}) = \{y\} \quad C_{a_4}(\{z\}) = \{z\}$$
$$C_{a_1}(\{x'\}) = \{x'\}$$
$$C_{a_1}(\{x, x'\}) = \{x\}$$

	$C_{a_3}(\{x, x'\}) = \{x\}$	$C_{a_3}(\{x, x', y\}) = \varnothing$	$C_{a_3}(\{x, x', y, z\}) = \{x, x', y\}$
$C_{a_3}(\{x\}) = \{x\}$	$C_{a_3}(\{x, y\}) = \{y\}$	$C_{a_3}(\{x, x', z\}) = \{x, x'\}$	
$C_{a_3}(\{x'\}) = \{x'\}$	$C_{a_3}(\{x', y\}) = \{x'\}$	$C_{a_3}(\{x, y, z\}) = \{x, y\}$	
$C_{a_3}(\{y\}) = \{y\}$	$C_{a_3}(\{x, z\}) = \{x\}$	$C_{a_3}(\{x', y, z\}) = \{x', y\}$	
$C_{a_3}(\{z\}) = \varnothing$	$C_{a_3}(\{x', z\}) = \{x'\}$		
	$C_{a_3}(\{y, z\}) = \{y\}$		

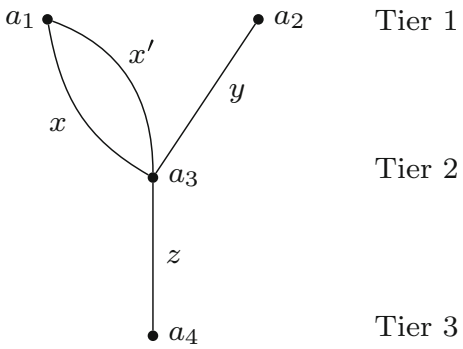

Fig. 1 The set of all possible contracts in Example A

Choice functions satisfy both same-side substitutes and cross-side complements. In particular, note that $U_{a_3}(\{x, x', y\}) = U_{a_3}(\{x, x', y, z\}) = \{x, x', y\}$ and $D_{a_3}(\{x, x', y\}) = \varnothing \subset D_{a_3}(\{x, x', y, z\}) = \{z\}$. Then a_3's choice functions satisfy cross-side complements, since

$$\varnothing = U_{a_3}(C_{a_3}(\{x, x', y\})) \subset U_{a_3}(C_{a_3}(\{x, x', y, z\})) = \{x, x', y\}.$$

Moreover, they satisfy same-side substitutes since

$$\varnothing = D_{a_3}(\{x, x', y\}) \backslash D_{a_3}(C_{a_3}(\{x, x', y\})) \subset D_{a_3}(\{x, x', y, z\}) \backslash D_{a_3}(C_{a_3}(\{x, x', y, z\})) = \{z\},$$

It is easy to verify the same for each node.

However, contracts for a_3 do not satisfy the IRC condition, since $z \notin C_{a_3}(\{x, x', y, z\})$ and yet

$$\varnothing = C_{a_3}(\{x, x', y\}) \neq C_{a_3}(\{x, x', y, z\} = \{x, x', y\}.$$

No chain stable network exists in this example.

Network	Chain block	Network	Chain block
$\{x\}$	$\{y\}$	$\{x, y\}$	$\{y\}$
$\{x'\}$	$\{x\}$	$\{x', y\}$	$\{x'\}$
$\{y\}$	$\{x'\}$	$\{x, z\}$	$\{x\}$
$\{z\}$	$\{x\}$	$\{x', z\}$	$\{x'\}$
$\varnothing$	$\{x\}$	$\{x, y, z\}$	$\{x, y\}$
		$\{x', y, z\}$	$\{x', y\}$

Notice that if $A = \{a_1, a_2, a_3\}$, we are left with two tiers and the example is the same as Example 1 in Aygün and Sönmez (2012b).

Appendix B: An example for the operation of the supply chain cumulative offer algorithm

Consider $A = A_1 \cup A_2 \cup A_3 = \{a\} \cup \{b_1, b_2\} \cup \{c\}$ with contracts $\mathcal{X} = \{x, y, z, w\}$ where

$$x = (a, b_1), y = (a, b_2), z = (b_1, c), w = (b_2, c).$$

Figure 2 illustrates all of the possible contracts that can be signed. Choice functions are as follows:

$C_a(\{x, y\}) = \{x\}$	$C_{b_1}(\{x, z\}) = \{x, z\}$	$C_c(\{w\}) = \{w\}$
$C_a(\{x\}) = \{x\}$	$C_{b_1}(X) = \emptyset, \forall X \subseteq \mathcal{X} \backslash \{x, z\}$	$C_c(X) = \emptyset, \forall X \subseteq \mathcal{X} \backslash \{w\}$
$C_a(\{y\}) = \{y\}$	$C_{b_2}(\{y, w\}) = \{y, w\}$	
$C_a(X) = \emptyset, \forall X \subseteq \mathcal{X} \backslash \{x, y\}$	$C_{b_1}(X) = \emptyset, \forall X \subseteq \mathcal{X} \backslash \{x, z\}$	

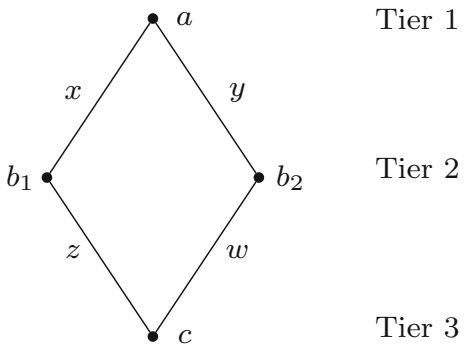

Fig. 2 The set of all possible contracts in Example B

The supply chain cumulative offer algorithm (SCCOA) proceeds as follows. Initialize the opportunity sets as $O^U_{\tilde{a}}(1) = \emptyset, \forall \tilde{a} \in A$ and

$$\begin{aligned} O^D_a(1) &= \{x, y\} \\ O^D_{b_1}(1) &= \{z\} \\ O^D_{b_2}(1) &= \{w\} \\ O^D_c(1) &= \emptyset \end{aligned}$$

Let $t := 1$ and suppose a makes the first offer,

$$Y(1) = D_a(C_a(O^U_a(1) \cup O^D_a(1))) = D_a(C_a(\emptyset \cup \{x, y\})) = \{x\}$$

Then $Y_{b_1}(1) = \{x\}$ and $Y_{\tilde{a}}(1) = \emptyset$ for all $\tilde{a} \in A \backslash \{a, b_1\}$. The held sets are thus

$$H_{b_1}(1) = U_{b_1}(C_{b_1}(O^U_{b_1} \cup \{x\} \cup \{z\})) = \{x\}$$

and $H_{\tilde{a}}(1) = \emptyset$ for all $\tilde{a} \in A \backslash \{b_1\}$.

Now update the opportunity sets. Since b_1 is the only agent that receives an offer, $O^U_{b_1}(t) = \{x\}$ and $O^U_{\tilde{a}}(t) = O^U_{\tilde{a}}(t-1) = \emptyset$ for all $\tilde{a} \in A \backslash \{b_1\}$. Since no offers have been rejected, the downstream opportunity sets remain the same; that is, $O^D_{\tilde{a}}(t) = O^D_{\tilde{a}}(t-1)$ for all $\tilde{a} \in A$.

Let $t := 2$ and suppose b_1 makes the next offer (the reader can verify that the allocation is the same regardless of who makes the offer). Then b_1 offers

$$Y(2) = D_{b_1}(C_{b_1}(O^U_{b_1}(2) \cup O^D_{b_1}(2))) = D_{b_1}(C_{b_1}(\{x\} \cup \{x, z\})) = \{z\}.$$

Then $Y_c(2) = \{z\}$ and $Y_{\tilde{a}}(2) = \emptyset$ for all other agents. The held sets are

$$H_c(2) = U_c(C_c(O^U_c(2) \cup \{z\} \cup O^D_c(2))) = \emptyset,$$

that is, c rejects the offer from b_1, and $H_{\tilde{a}}(2) = H_{\tilde{a}}(1)$ for all other agents $\tilde{a}$.

Now update the opportunity sets. The only upstream opportunity set that is updated is $O_c^U(t) = \{z\}$, since c is the only agent that receives an offer. Since b_1's offer was rejected at $t-1$, b_1's downstream opportunity set is $O_{b_1}^D(t) = \emptyset$ and $O_{\tilde{a}}^D(t) = O_{\tilde{a}}^D(t-1)$ for all other agents.

Let $t := 3$ and suppose b_1 is chosen to make another offer. Since $O_{b_1}^D = \emptyset$, agent b_1 cannot make an offer. So we pick another agent to make an offer, say b_2. Then b_1 offers

$$Y(3) = D_{b_1}(C_{b_1}(O_{b_1}^U(3) \cup O_{b_1}^D(3))) = D_{b_1}(C_{b_1}(\emptyset \cup \{w\})) = \{w\}.$$

Now agent c accepts the offer, $H_c(3) = \{w\}$. Moreover, since b_1's downstream opportunity set has changed, the set of contracts that b_1 holds also changes:

$$H_{b_1}(3) = U_{b_1}(C_{b_1}(\{x\} \cup \emptyset)) = \emptyset$$

Thus, c's rejection of b_1's offer triggers a rejection of a's previously held downstream offer to b_1. For the remaining agents, a, b_2, their set of held contracts remains the same.

Now update the opportunity sets. Since c receives a new offer so $O_c^U(t) = \{z, w\}$, while $O_{\tilde{a}}^U(t) = O_{\tilde{a}}^U(t-1)$ for all other agents. On the other hand, a's previously held offer is no longer being held by b_1, a's downstream opportunity set shrinks, $O_a^D(5) = \{y\}$, while $O_{\tilde{a}}^D(t) = O_{\tilde{a}}^D(t-1)$ for all other agents.

Let $t := 4$ and suppose a makes an offer,

$$Y(4) = D_a(C_a(O_a^U(4) \cup O_a^D(4))) = \{y\}$$

We have $Y_{b_2} = \{y\}$ and $Y_{\tilde{a}} = \emptyset$ for the remaining agents. Agent b_2 accepts, so the held sets are $H_{b_2}(4) = \{y\}$ and $H_{\tilde{a}}(4) = H_{\tilde{a}}(3)$ for all other agents.

Now update the opportunity sets. Since b_2 gets the sole offer, $O_{b_2}^U(t) = \{y\}$ and $O_{\tilde{a}}^U(t) = O_{\tilde{a}}^U(t-1)$ for all other agents. No contracts have been rejected downstream since $\{x, z\}$ were rejected, so $O_{\tilde{a}}^D(t) = O_{\tilde{a}}^D(t-1)$ for all agents.

Let $t := 5$. Now, a and b_2 have offers held and have no new offers to make. In addition, $O_{b_1}^D(t) = O_c^D(t) = \emptyset$, so neither b_1 or c have offers to make. Thus, $H_{\tilde{a}}(t) = H_{\tilde{a}}(t-1)$, for all $\tilde{a} \in A$, and the algorithm terminates at step $T := 5$.

The SCCOA terminates at this step since no downstream proposals are made and no upstream offers are rejected. The final allocation is $\mu = \cup_{\tilde{a} \in A} H_{\tilde{a}}(T) = \{y, w\}$. The reader can easily verify that this allocation is a chain stable network.

Appendix C: Proofs

Lemmas

We introduce the lemmas that will be used in the proof of Theorem 1.

Lemma 1 *Consider an agent $a \in A$ who receives a proposal of contract z at step $t < T$ in the algorithm. If agent a accepts the contract z at the last step T, then agent a does not reject the contract at any step t' for $t \leq t' \leq T$.*

Proof Suppose agent a is proposed with a contract z at step $t < T$. By construction of the algorithm, $z \in O_a^U(t) \subseteq O_a^U(T)$. Note that

$$z \in O_a^U(t) \cup O_a^D(T) \subseteq O_a^U(t) \cup O_a^D(t), \tag{1}$$

and

$$z \in O_a^U(t) \cup O_a^D(T) \subseteq O_a^U(T) \cup O_a^D(T). \tag{2}$$

We first show that agent a accepts contract z at step t.

Suppose agent a rejects contract z at step t and accepts contract z at step T. Since agent a rejects contract z at step t, $z \notin U_a\Big[C_a\Big(O_a^U(t) \cup O_a^D(t)\Big)\Big]$. Cross-side complements and Eq. (1) imply that

$$z \notin U_a\Big[C_a\Big(O_a^U(t) \cup O_a^D(T)\Big)\Big].$$

Moreover, same-sided substitutes and Eq. (2) imply that

$$z \notin \Big[C_a\Big(O_a^U(T) \cup O_a^D(T)\Big)\Big].$$

This is a contradiction to the supposition that agent a accepts contract z at step T. Therefore, agent a accepts contract z at step t.

We now show that agent a does not reject contract z at any step t' for $t \leq t' \leq T$.

We've already shown that agent a accepts contract z at step t. Consider a step t' with $t \leq t' \leq T$. Cross-side complements imply

$$z \in O_a^U(T) \cup O_a^D(T) \subseteq O_a^U(T) \cup O_a^D(t').$$

Therefore, $z \in U_a\Big[C_a\Big(O_a^U(T) \cup O_a^D(t')\Big)\Big]$. Same-sided substitutes imply

$$z \in O_a^U(t') \cup O_a^D(t') \subseteq O_a^U(T) \cup O_a^D(T).$$

Therefore, $z \in U_a\Big[C_a(O_a^U(t') \cup O_a^D(t'))\Big]$. Thus, agent a holds contract z at every step t' for $t \leq t' \leq T$.

Lemma 2 *Let μ be the allocation produced by the algorithm. We have $D_a(\mu(a)) = D_a(C_a(\mu))$ for all $a \in A$.*

Proof The statement is immediately true if $a \in A_K$. In the following, we consider an agent $a \in A \backslash A_K$. Note that $D_a(C_a(\mu(a))) = D_a(C_a(\mu)) \subseteq D_a(\mu) = D_a(\mu(a))$. The first equality is by IRC, and the second one is by definition.

We show that there does not exist a contract $z \in D_a(\mathcal{X})$ such that $z \in D_a(\mu)$ and $z \notin D_a(C_a(\mu))$.

Suppose, on the contrary, that such a contract z exists, i.e., there exists $z \in D_a(\mathcal{X})$ such that $z \in D_a(\mu)$ and $z \notin D_a(C_a(\mu))$. Let $b \equiv B(z)$ be the buyer associated with contract z. Then, $z \in \mu$ means that agent a proposes contract z to agent b at some step and that agent b holds contract z at step T. Let $t < T$ be the step at which agent a proposes contract z to agent b. By Lemma 1, agent b holds contract z at step t and step T. Therefore, $z \in O_a^D(T) \subseteq O_a^D(t)$. Note that

$$z \in O_a^U(t) \cup O_a^D(T) \subseteq O_a^U(t) \cup O_a^D(t), \tag{3}$$

and

$$z \in O_a^U(t) \cup O_a^D(T) \subseteq O_a^U(T) \cup O_a^D(T). \tag{4}$$

The fact that agent a proposes z to agent b at step t implies $z \in D_a\Big[C_a\Big(O_a^U(t) \cup O_a^D(t)\Big)\Big]$. Together with same-sided substitutes and Eq. (3), this implies $z \in D_a\Big[C_a\Big(O_a^U(t) \cup O_a^D(T)\Big)\Big]$. Moreover, together with cross-side complements and equation (4), this further implies that $z \in D_a\Big[C_a\Big(O_a^U(T) \cup O_a^D(T)\Big)\Big]$.

On the other hand, $z \notin D_a(C_a(\mu))$ means $z \notin D_a\Big[C_a\Big(O_a^U(T) \cup O_a^D(T)\Big)\Big]$. This is a contradiction. Therefore, such z does not exist.

Lemma 3 *Let μ be the allocation produced by the algorithm. We have $U_a(\mu(a)) = U_a(C_a(O_a^U(T)) \cup O_a^D(T))$ for all $a \in A$.*

Proof The statement is immediately true if $a \in A_1$. In the following, we consider an agent $a \in A \backslash A_1$. Note that $U_a(\mu) = U_a(\mu(a))$ by definition, and $U_a(C_a(\mu(a))) = U_a(C_a(\mu))$ by IRC.

Moreover, $U_a[U_b(C_b(O_b^U(T) \cup O_b^D(T)))] = \emptyset$ for all $b \in A \backslash \{a\}$. Hence,

$$\begin{aligned} U_a(\mu) &= U_a[\bigcup_{b \in A} U_a(C_a(O_a^U(T) \cup O_a^D(T)))] \\ &= U_a(C_a(O_a^U(T) \cup O_a^D(T))). \end{aligned}$$

□

Proof of Theorem 1

We show that the cumulative offer algorithm produces a chain stable network.

First, note that by construction of the algorithm, any rejected contract is not proposed again in the algorithm. Since the set of contracts is finite, the algorithm terminates at a finite step.

Next, we show that the allocation produced by the algorithm, μ, is a chain stable network. There are two steps.

Step 1 μ is individually rational.

Suppose $C_a(\mu(a)) \neq \mu(a)$ for some agent $a \in A$. By Lemma 2, $D_a(\mu(a)) = D_a(C_a(\mu))$. Then, this implies $U_a(C_a(\mu(a))) \neq U_a(\mu(a))$. Since $U_a(C_a(\mu(a))) \subseteq U_a(\mu(a))$, this further implies that there exists a contract $x \in \mu$ such that

$$x \in U_a(\mu(a)) \text{ and } x \notin U_a(C_a(\mu(a))). \tag{5}$$

We show that x does not exist.

By Lemma 3, $U_a(\mu(a)) = U_a(C_a(O_a^U(T) \cup O_a^D(T))$. Thus, Eq. (5) is equivalent to

$$x \in U_a(C_a(O_a^U(T) \cup O_a^D(T))) \text{ and } x \notin U_a[C_a(C_a(O_a^U(T) \cup O_a^D(T)))] \tag{6}$$

Consider upstream contracts for a:

$$\begin{aligned}
&U_a\Big[C_a\Big\{C_a(O_a^U(T) \cup O_a^D(T))\Big\}\Big] \\
&\quad = U_a\Big[C_a\Big\{U_a\Big(C_a(O_a^U(T) \cup O_a^D(T))\Big) \cup D_a\Big(C_a(O_a^U(T) \cup O_a^D(T))\Big)\Big\}\Big] \\
&\quad \subseteq U_a\Big[C_a\Big\{U_a\Big(C_a(O_a^T(T) \cup O_a^D(T))\Big) \cup D_a\Big(O_a^U(T) \cup O_a^D(T)\Big)\Big\}\Big] \\
&\quad \subseteq U_a\Big[C_a\Big\{U_a\Big(O_a^U(T) \cup O_a^D(T)\Big) \cup D_a\Big(O_a^U(T) \cup O_a^D(T)\Big)\Big\}\Big] \\
&\quad = U_a\Big[C_a\Big\{O_a^U(T) \cup O_a^D(T)\Big\}\Big]
\end{aligned} \tag{7}$$

Note that the first inclusion holds because of cross-sided complements. Equation (7) contradicts Eq. (6). Therefore, μ is individually rational.

Step 2 μ is not blocked by any chain.

Suppose μ is blocked by the chain $\mathbf{z} = \{z_1, \ldots, z_n\}$. Let $b_i \equiv B(z_i)$ for $i = 1, \ldots, n$. Consider agent b_n and define the following.

1. $X'_{b_n} \equiv \mu(b_n)$: the set of contracts held by b_n in μ.
2. $X''_{b_n} \equiv C_{b_n}(\mu(b_n) \cup \{z_n\})$: the new set of contracts held by b_n.

Then, $z_n \notin \mu$ and $z_n \in C_{b_n}(\mu(b_n) \cup \{z_n\})$ imply

$$U_{b_n}[X''_{b_n}] \neq U_{b_n}[C_{b_n}(X'_{b_n})]. \tag{8}$$

By construction of the algorithm, $U_{b_n}[X''_{b_n}] \subseteq O_{b_n}^U(T)$, since the contract in the blocking chain is proposed prior to step T. Moreover, $D_{b_n}[X''_{b_n}] \subseteq O_{b_n}^D(T)$, since b_n

does not take on new downstream contracts.[4] So, $X''_{b_n} \subseteq O^U_{b_n}(T) \cup O^D_{b_n}(T)$. Therefore,

$$\begin{aligned} U_{b_n}[X''_{b_n}] &= U_{b_n}[C_{b_n}(X'_{b_n} \cup X''_{b_n})] && (9)\\ &= U_{b_n}[C_{b_n}(O_{b_n}(T) \cup D_{b_n}(T) \cup X''_{b_n})] && (10)\\ &= U_{b_n}[C_{b_n}(O_{b_n}(T) \cup D_{b_n}(T))] && (11)\\ &= U_{b_n}[C_{b_n}(X'_{b_n})], && (12) \end{aligned}$$

contradicting Eq. (8). Note that

1. Equality (9) holds because $U_{b_n}[X''_{b_n}] = U_{b_n}[C_{b_n}(X''_{b_n})]$ and $U_{b_n}[C_{b_n}(X_{b_n})''] = U_{b_n}[C_{b_n}(X'_{b_n} \cup X''_{b_n})]$.
2. Equality (10) holds because

$$X'_{b_n} = C_{b_n}(X'_{b_n}) = C_{b_n}(O^U_{b_n}(T) \cup O^D_{b_n}(T)). \tag{13}$$

3. Equality (11) holds by Lemma 1, since $C_{b_n}(O^U_{b_n}(T) \cup O^D_{b_n}(T)) \cup X''_{b_n} \subseteq O^U_{b_n}(T) \cup O^D_{b_n}(T)$.
4. Equality (12) holds because Eq. (13) implies

$$U_{b_n}[X'_{b_n}] = U_{b_n}[C_{b_n}(X'_{b_n})] = U_{b_n}[C_{b_n}(O^U_{b_n}(T) \cup O^D_{b_n}(T))].$$

Thus, μ is a chain stable network.

Proof of Proposition 1

Consider $Y \subset \mathcal{X}$ and $z \in \mathcal{X} \backslash Y$ such that $z \notin C(Y \cup \{z\})$. If $z \in \mathcal{X} \backslash X_a$, then $U(Y) = U(Y \cup \{z\})$ and $D(Y) = D(Y \cup \{z\})$. Thus, by same-side substitutes and cross-side complements, we can show $C(Y) = C(Y \cup \{z\})$.

In the following, assume that $z \in X_a$ (without loss of generality assume $z \in U_a$). Then, $U(Y) \subset U(Y \cup \{z\})$ and $D(Y) = D(Y \cup \{z\})$. Thus, same-sided substitutes implies $U(C(Y \cup \{z\}))$. Therefore, $|U(C(Y \cup \{z\}))| \leq |U(C(Y))|$ and $|D(C(Y))| \leq |D(C(Y \cup \{z\}))|$. On the other hand, by the Law of Aggregate Demand, we have

$$0 \geq |U(C(Y \cup \{z\}))| - |U(C(Y))| \geq |D(C(Y \cup \{z\}))| - |D(C(Y))| \geq 0$$

Thus, $|U(C(Y \cup \{z\}))| = |U(C(Y))|$ and $|D(C(Y))| = |D(C(Y \cup \{z\}))|$. We therefore have $U(C(Y)) = U(C(Y \cup \{z\}))$, $D(C(Y)) = D(C(Y \cup \{z\}))$, and $C(Y) = C(Y \cup \{z\})$. This implies that choice functions satisfy the IRC condition.

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[4] Note that b_n is the agent in the blocking chain who is in the lowest tier.

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